

Assignment 5

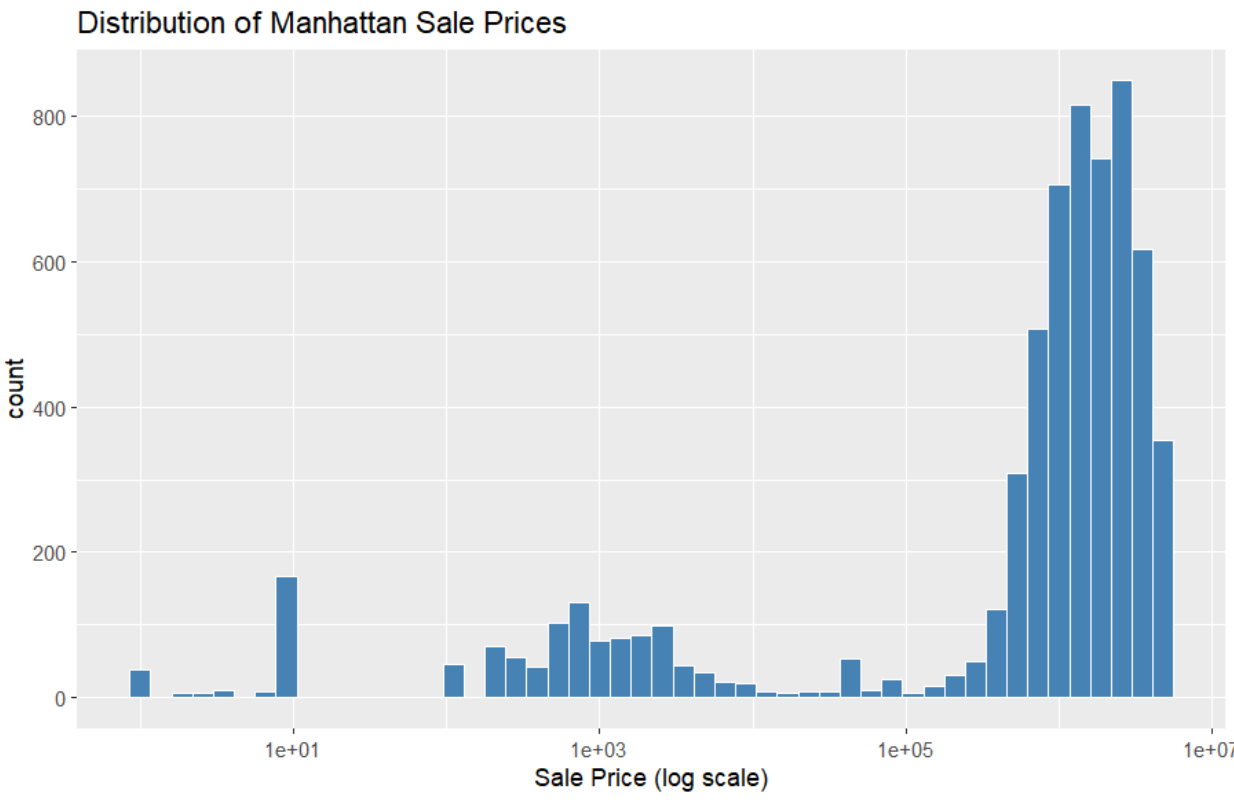
1.

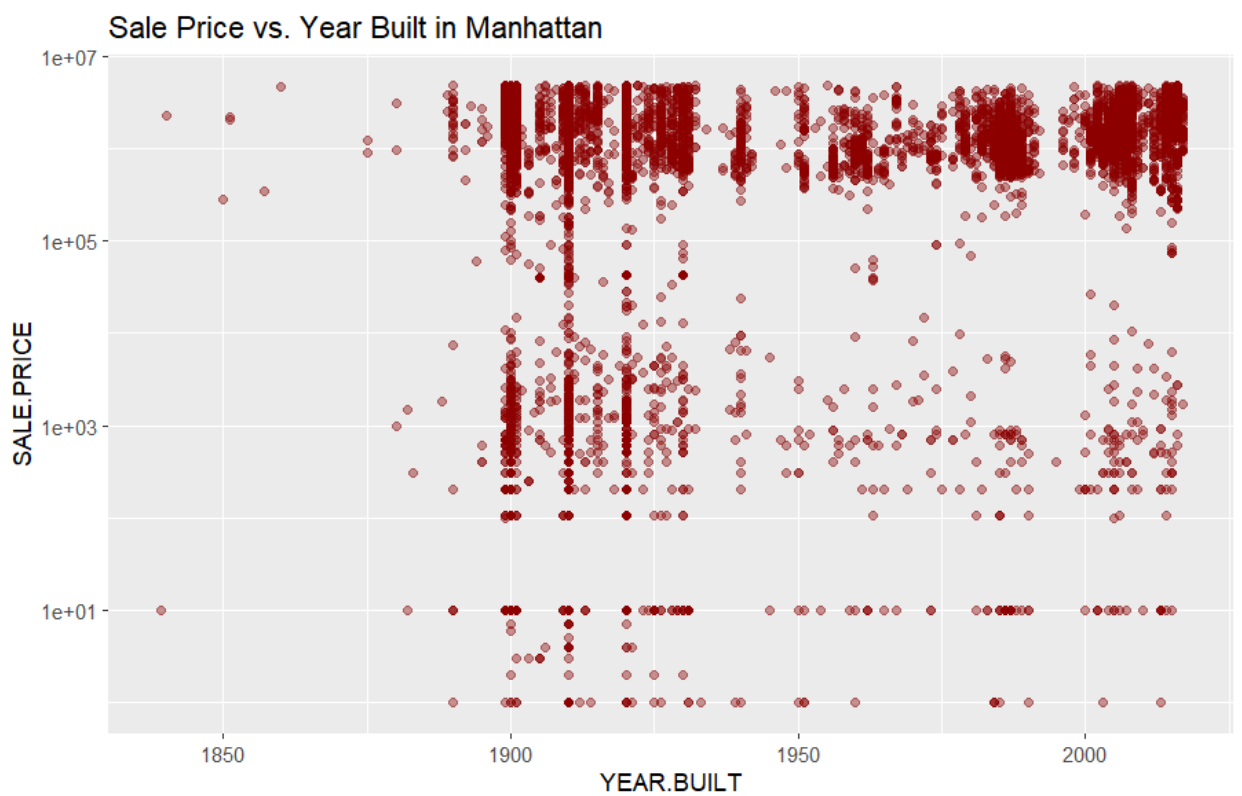
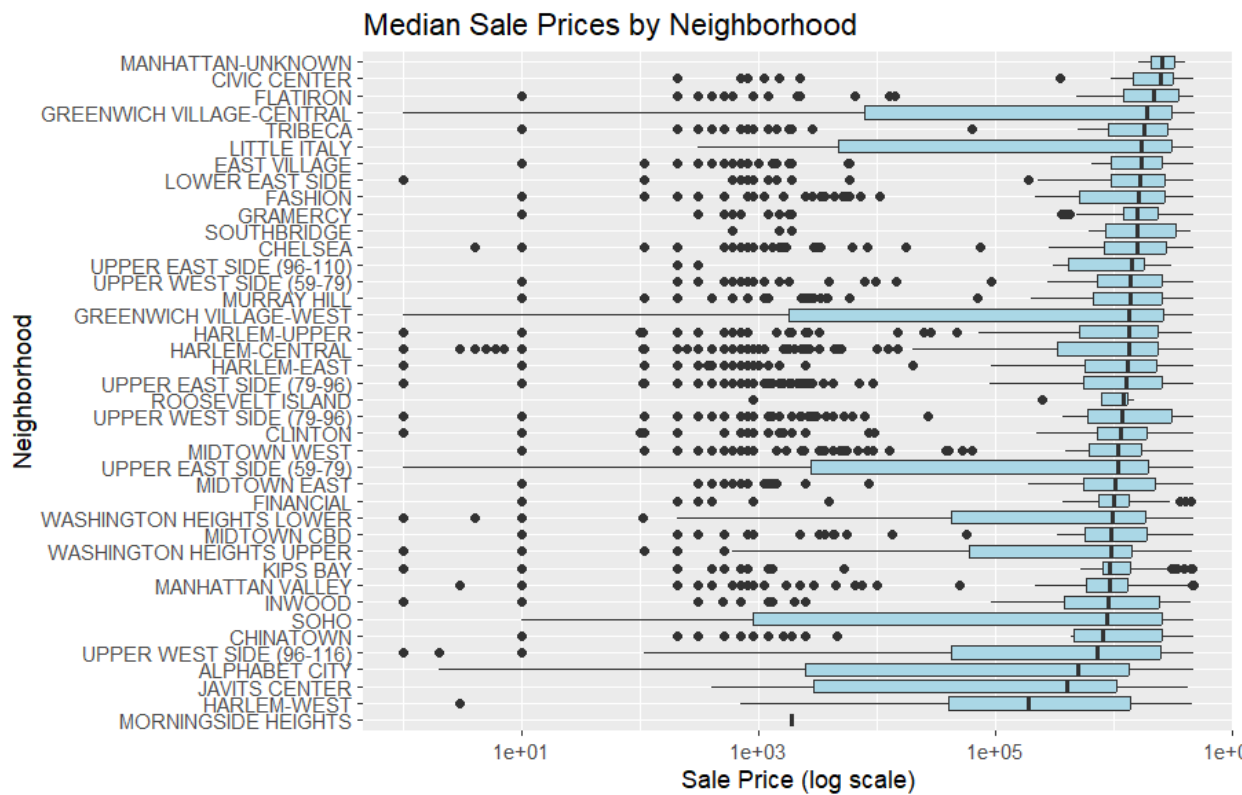
- a. I would look for relationships between sale price and building/lot characteristics (gross square feet, land square feet, year built, and unit counts), spatial patterns (latitude/longitude clusters of high vs low prices), and seasonal or tax-class effects. I'll explore these with summary statistics, histograms and boxplots for distributions, scatterplots and maps for spatial patterns, and correlation/PCA to reduce multicollinearity before modeling. For modeling I'll try simple univariate regressions to screen features, then multiple linear regression (and, if needed, log-transforming price) and tree-based models to capture nonlinearity.
- b. I examined variable summaries (min/1stQ/median/mean/3rdQ/max) for SALE.PRICE, YEAR.BUILT, and GROSS.SQUARE.FEET to understand central tendency and spread; for Manhattan your summary shows a median sale price around \$1.26M and a mean larger than the median, indicating a right skew. I plotted histograms and boxplots for these variables and used scatterplots of SALE.PRICE vs GROSS.SQUARE.FEET and SALE.PRICE vs YEAR.BUILT to inspect relationships and extreme points. Outliers are clear in SALE.PRICE — there are very large prices (max in the multi-million range) that drive the mean up and inflate residuals; these are visible compared with the 1st/3rd quartiles in your summaries. To demonstrate outliers I used boxplots and residuals-vs-fitted plots for univariate linear models. I also noted missingness in some predictors (observations deleted in regressions), so I recorded how many rows were lost when particular features had NAs.

```
> summary(manhattan$SALE.PRICE)
   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
    1    530000 1260000 1530382 2369575 4818571

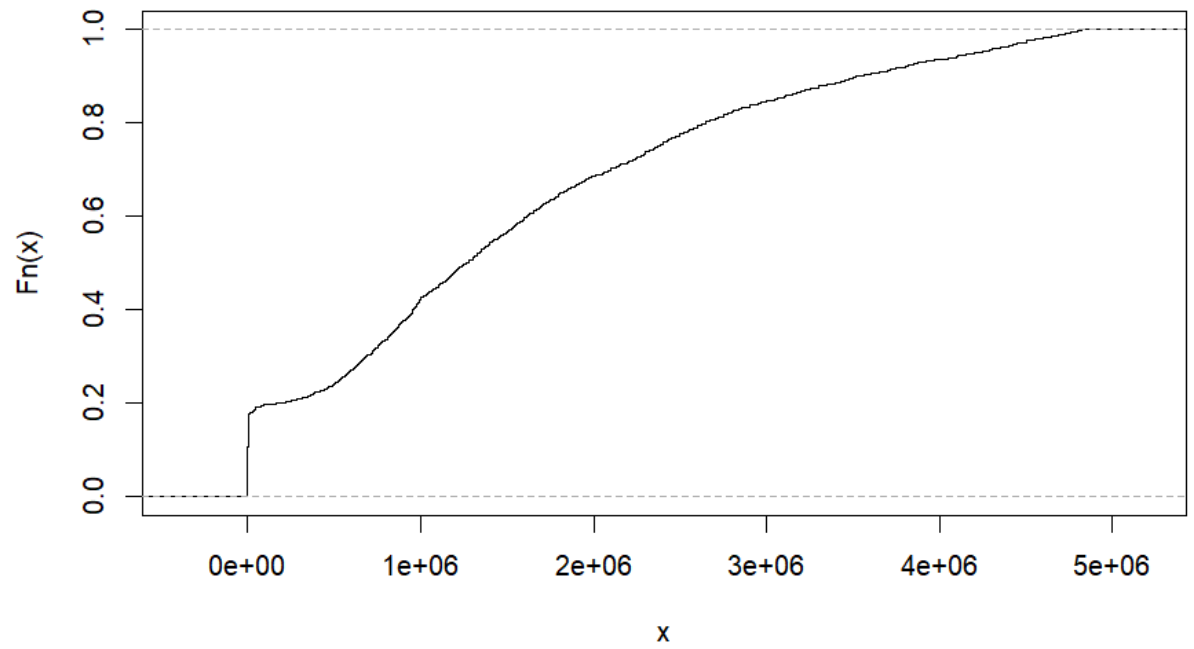
> summary(manhattan$YEAR.BUILT)
   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
```

```
1839    1910    1938    1952    2002    2017
> summary(manhattan$GROSS.SQUARE.FEET)
Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
 207    813    1497    5173   4200 112493
```

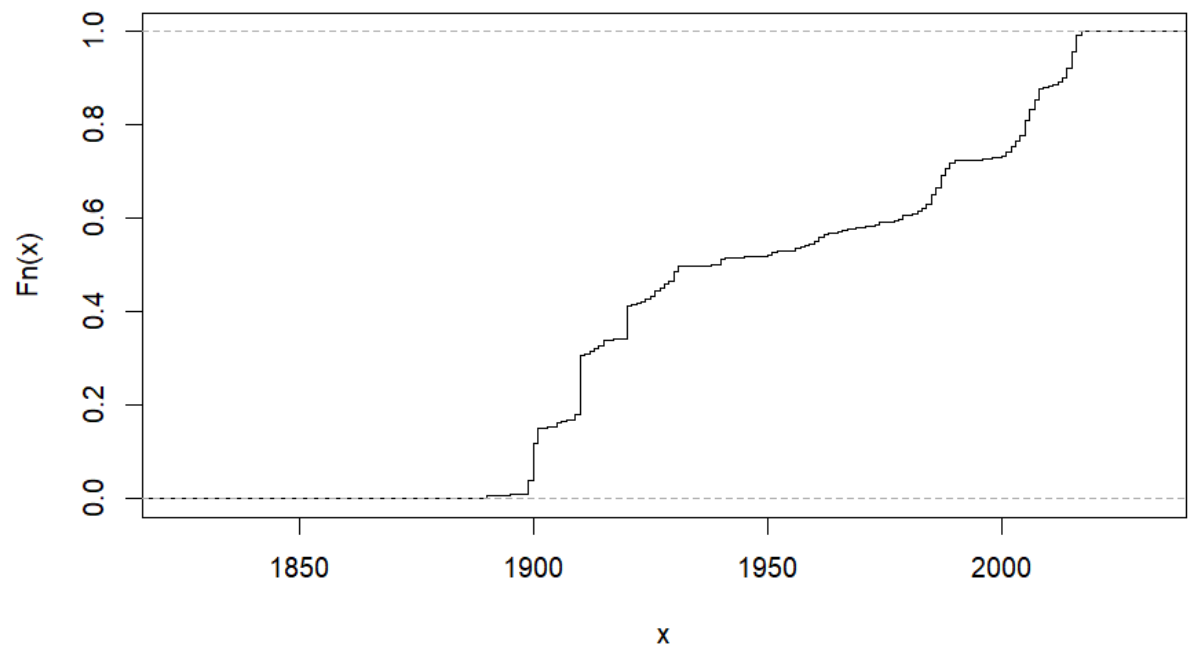


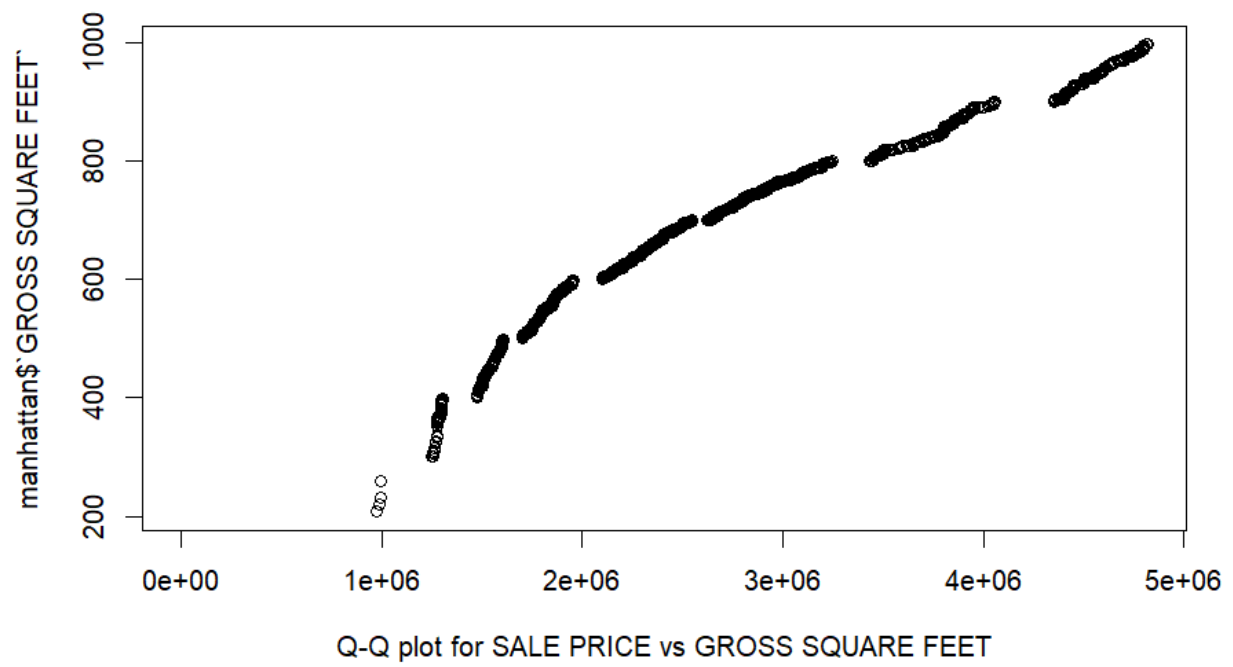
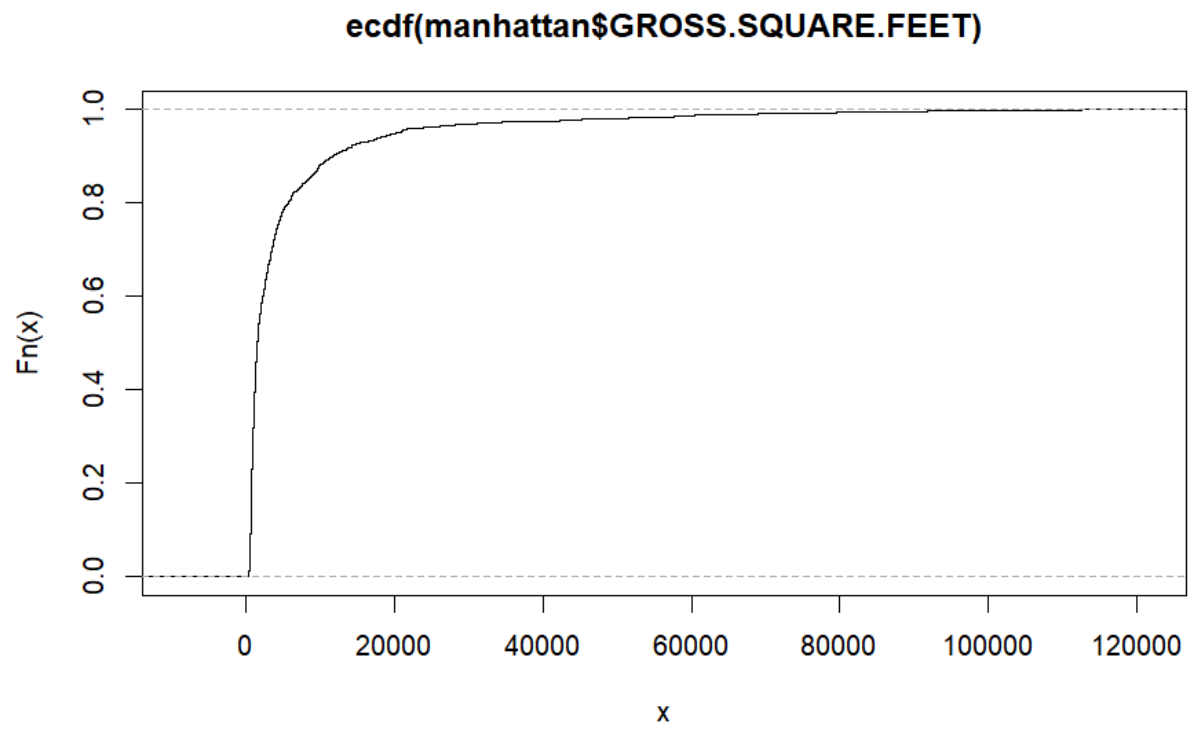


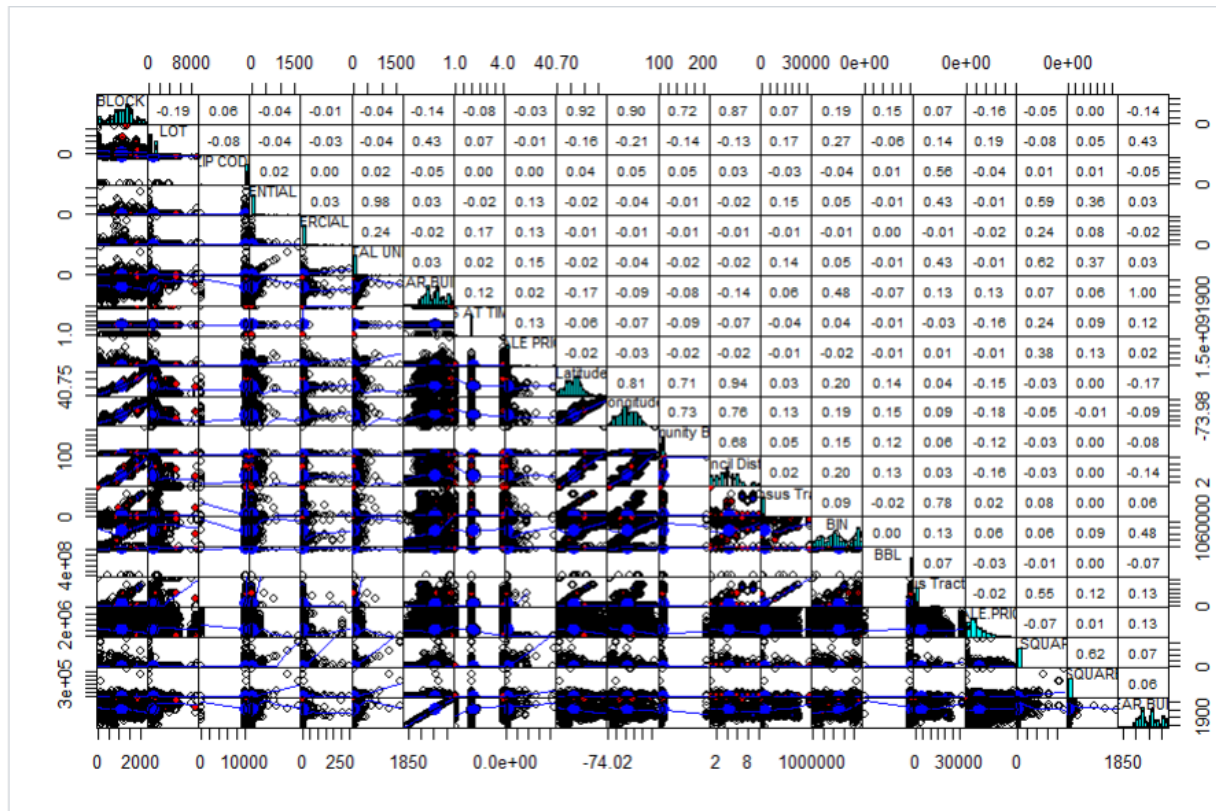
ecdf(manhattan\$SALE.PRICE)



ecdf(manhattan\$YEAR.BUILT)







- c. I ran univariate regressions for candidate predictors and used PCA to identify high-variance contributors; YEAR.BUILT, Latitude, Longitude, and GROSS.SQUARE.FEET emerged as the most useful predictors from PCA and single-variable regressions. In Manhattan, GROSS.SQUARE.FEET produced the largest univariate R^2 you reported (~ 0.029) and was highly significant ($p < 2e-16$), while YEAR.BUILT, latitude and longitude were also statistically significant but each explained only a tiny portion of variance (adjusted R^2 values near zero to a few hundredths). Because R^2 remains low, I cleaned data by removing or flagging extreme SALE.PRICE outliers, and I handled missing predictor values (rows with NA were excluded in `lm` calls), and I considered transforming SALE.PRICE (e.g., log) or using interactions/regularization or tree-based methods to improve fit. In short, simple linear models explain only a small fraction of Manhattan price variance — suggesting important omitted variables (condition, interior features, building class, transaction type) or strong nonlinearity. After running regression analysis below, we can see that after

evaluating these models, the YEAR BUILT, Longitude, Latitude, and GROSS SQUARE FEET features are the best models compared to the rest of them show weak correlation.

```
> principal_components$loadings
```

Loadings:

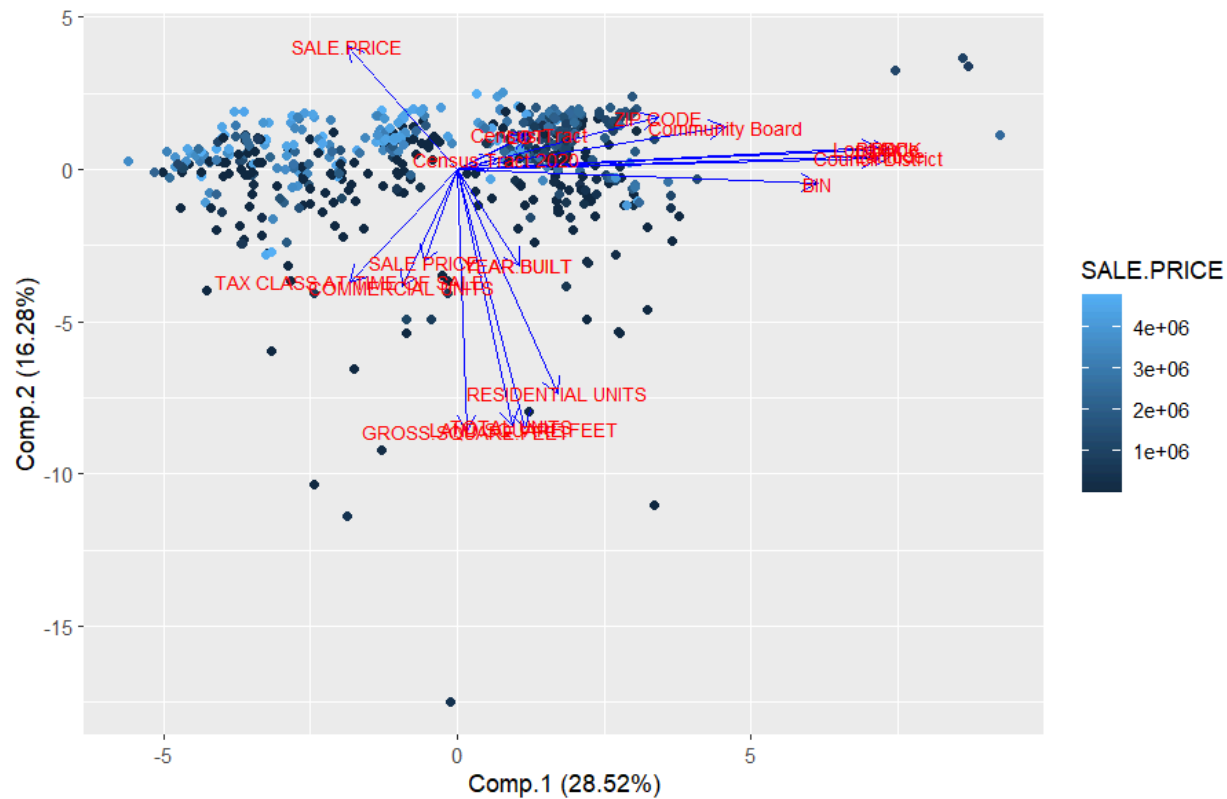
	Comp.1	Comp.2	Comp.3	Comp.4	Comp.5	Comp.6	Comp.7
Comp.8 Comp.9							
BLOCK	0.391				0.129		
LOT			-0.284	-0.210	-0.115	-0.365	-0.100
0.371 -0.287							
ZIP CODE	0.181		-0.185		-0.645	0.107	
RESIDENTIAL UNITS		-0.391	0.211			-0.159	-0.421
0.169							
COMMERCIAL UNITS		-0.205		0.163	-0.181	-0.346	0.657
0.104 -0.300							
TOTAL UNITS		-0.447	0.190		-0.141	-0.327	
YEAR BUILT		-0.167	-0.544	-0.202	0.145	-0.106	
-0.159							
TAX CLASS AT TIME OF SALE		-0.196	-0.177	0.114		0.232	0.495
-0.316 0.332							
SALE PRICE		-0.161	-0.207			0.589	
0.596 -0.227							
Latitude	0.392						
Longitude	0.380						
Community Board	0.242		-0.129		-0.596	0.115	
Council District	0.380				0.129		
Census Tract			-0.194	0.628			-0.198
BIN	0.326				0.195	-0.193	
-0.140							
BBL	0.391				0.129		
Census Tract 2020			-0.222	0.647			-0.122
SALE.PRICE		0.215				-0.269	0.133
0.530 0.734							
GROSS.SQUARE.FEET		-0.457				0.166	
0.195							
LAND.SQUARE.FEET		-0.451					-0.122
0.224							
YEAR.BUILT		-0.167	-0.544	-0.202	0.145	-0.106	
-0.159							
	Comp.10	Comp.11	Comp.12	Comp.13	Comp.14	Comp.15	
Comp.16 Comp.17							
BLOCK					0.368	0.116	
0.192 0.187							
LOT	0.686		0.103				
ZIP CODE			-0.155		0.178	0.377	
-0.503							

RESIDENTIAL UNITS		-0.405	0.168				
COMMERCIAL UNITS	-0.217	0.160	0.116				
TOTAL UNITS	-0.155	-0.257	0.208				
YEAR BUILT	-0.217			0.105			
TAX CLASS AT TIME OF SALE	0.449	-0.439					
SALE PRICE	-0.207	-0.279	0.169	-0.123			
Latitude					0.270	-0.188	
-0.133							
Longitude			0.229		-0.576	0.589	
-0.181 0.219							
Community Board					-0.160	-0.490	
0.514							
Council District			0.203	0.155	-0.262	-0.403	
-0.270 -0.569							
Census Tract			0.112		-0.280	-0.125	
0.612							
BIN	-0.124	-0.340	-0.734	-0.251	-0.219		
BBL					0.368	0.116	
0.192 0.187							
Census Tract 2020					0.239		
-0.653 0.129							
SALE.PRICE	-0.148						
GROSS.SQUARE.FEET	0.140	0.306	-0.419	0.636			
LAND.SQUARE.FEET	0.203	0.471		-0.660			
YEAR.BUILT	-0.217			0.105			
	Comp.18	Comp.19	Comp.20	Comp.21			
BLOCK	0.292	0.707					
LOT							
ZIP CODE							
RESIDENTIAL UNITS			0.594				
COMMERCIAL UNITS			0.385				
TOTAL UNITS			-0.696				
YEAR BUILT				-0.702			
TAX CLASS AT TIME OF SALE							
SALE PRICE							
Latitude	-0.824						
Longitude	-0.126						
Community Board							
Council District	0.350						
Census Tract							
BIN							
BBL	0.292	-0.707					
Census Tract 2020							
SALE.PRICE							
GROSS.SQUARE.FEET							
LAND.SQUARE.FEET							
YEAR.BUILT				0.702			

	Comp.1	Comp.2	Comp.3	Comp.4	Comp.5	Comp.6	Comp.7	Comp.8	Comp.9
Comp.10									
SS loadings	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
1.000									
Proportion Var	0.048	0.048	0.048	0.048	0.048	0.048	0.048	0.048	0.048
0.048									
Cumulative Var	0.048	0.095	0.143	0.190	0.238	0.286	0.333	0.381	0.429
0.476									

	Comp.11	Comp.12	Comp.13	Comp.14	Comp.15	Comp.16	Comp.17	Comp.18
Comp.19								
SS loadings	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
1.000								
Proportion Var	0.048	0.048	0.048	0.048	0.048	0.048	0.048	0.048
0.048								
Cumulative Var	0.524	0.571	0.619	0.667	0.714	0.762	0.810	0.857
0.905								

	Comp.20	Comp.21
SS loadings	1.000	1.000
Proportion Var	0.048	0.048
Cumulative Var	0.952	1.000



```
> for (feature in features){
```

```
+ formula <- as.formula(paste("SALE.PRICE ~", feature))
+ lin.mod <- lm(formula, data = manhattan_data)
+ print(feature)
+ print(summary(lin.mod))
+ }
[1] "YEAR.BUILT"
```

Call:

```
lm(formula = formula, data = manhattan_data)
```

Residuals:

	Min	1Q	Median	3Q	Max
	-1612293	-1004742	-281365	831489	3349923

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	-989397.3	717028.7	-1.380	0.167679
YEAR.BUILT	1290.7	367.2	3.515	0.000443 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1296000 on 6375 degrees of freedom

Multiple R-squared: 0.001934, Adjusted R-squared: 0.001778

F-statistic: 12.36 on 1 and 6375 DF, p-value: 0.0004426

```
[1] "SALE.PRICE"
```

Call:

```
lm(formula = formula, data = manhattan_data)
```

Residuals:

	Min	1Q	Median	3Q	Max
	-1530381	-1000382	-270382	839193	3288189

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	1530382	16242	94.22	<2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1297000 on 6376 degrees of freedom

```
[1] "Latitude"
```

Call:

```
lm(formula = formula, data = manhattan_data)
```

Residuals:

	Min	1Q	Median	3Q	Max
	-1679745	-995953	-272674	831759	3422427

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	96509660	17938435	5.380	7.72e-08 ***
Latitude	-2329686	440012	-5.295	1.23e-07 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1295000 on 6225 degrees of freedom

(150 observations deleted due to missingness)

Multiple R-squared: 0.004483, Adjusted R-squared: 0.004323

F-statistic: 28.03 on 1 and 6225 DF, p-value: 1.234e-07

[1] "Longitude"

Call:

lm(formula = formula, data = manhattan_data)

Residuals:

	Min	1Q	Median	3Q	Max
	-1718598	-989461	-277713	842883	3406301

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	-306761453	52396907	-5.855	5.03e-09 ***
Longitude	-4167682	708328	-5.884	4.22e-09 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1294000 on 6225 degrees of freedom

(150 observations deleted due to missingness)

Multiple R-squared: 0.005531, Adjusted R-squared: 0.005371

F-statistic: 34.62 on 1 and 6225 DF, p-value: 4.217e-09

[1] "BIN"

Call:

lm(formula = formula, data = manhattan_data)

Residuals:

	Min	1Q	Median	3Q	Max
	-1622889	-1018197	-272755	822245	3405108

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
--	----------	------------	---------	----------

```

(Intercept) -1.124e+06  6.021e+05  -1.866    0.062 .
BIN          2.519e+00  5.705e-01   4.415 1.03e-05 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

```

```

Residual standard error: 1295000 on 6223 degrees of freedom
(152 observations deleted due to missingness)
Multiple R-squared:  0.003123, Adjusted R-squared:  0.002962
F-statistic: 19.49 on 1 and 6223 DF,  p-value: 1.027e-05

```

```
[1] "BBL"
```

```

Call:
lm(formula = formula, data = manhattan_data)

```

```

Residuals:
      Min       1Q   Median       3Q      Max
-1544104  -993582  -268119   832867  3290190

```

```

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)  6.019e+05  5.198e+05   1.158   0.247
BBL          9.216e-04  5.140e-04   1.793   0.073 .
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

```

```

Residual standard error: 1297000 on 6223 degrees of freedom
(152 observations deleted due to missingness)
Multiple R-squared:  0.0005163, Adjusted R-squared:  0.0003557
F-statistic: 3.215 on 1 and 6223 DF,  p-value: 0.07303

```

```
[1] "GROSS.SQUARE.FEET"
```

```

Call:
lm(formula = formula, data = manhattan_data)

```

```

Residuals:
      Min       1Q   Median       3Q      Max
-1627965  -994970  -296650   799798  4305779

```

```

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)  1.633e+06  1.762e+04   92.65  <2e-16 ***
GROSS.SQUARE.FEET -1.976e+01  1.426e+00  -13.86  <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

```

Residual standard error: 1278000 on 6375 degrees of freedom
Multiple R-squared: 0.02925, Adjusted R-squared: 0.0291
F-statistic: 192.1 on 1 and 6375 DF, p-value: < 2.2e-16

- d. I trained k-NN, Random Forest, and Decision Tree models to classify NEIGHBORHOOD from quantitative predictors (price, gross area, etc.). Before training I encoded the neighborhood as a factor, scaled quantitative inputs for k-NN, and removed rows with missing predictor values; I also addressed class imbalance by stratified sampling for train/test. The k-NN results show many low per-class recall/precision values (sparse performance across many neighborhoods), while Random Forest gave much higher recall/precision/F1 for most neighborhoods (many near or above 0.95), indicating far superior performance on your Manhattan split. Decision Tree model report mostly NaN's and zeros. I evaluated with contingency tables plus per-class precision, recall and F1. The cleaning steps that mattered most were scaling for distance-based models and handling rare classes, since many neighborhoods have few examples and hurt both k-NN and overall metrics.

k-NN Model

```
> data.frame(recall, precision, f1)
```

	recall	precision	f1
ALPHABET CITY	0.25000000	0.28571429	0.26666667
CHELSEA	0.04225352	0.05000000	0.04580153
CHINATOWN	0.00000000	0.00000000	NaN
CIVIC CENTER	0.00000000	0.00000000	NaN
CLINTON	0.04000000	0.04255319	0.04123711
EAST VILLAGE	0.08333333	0.08888889	0.08602151
FASHION	0.08823529	0.10714286	0.09677419
FINANCIAL	0.09677419	0.08695652	0.09160305
FLATIRON	0.06060606	0.06666667	0.06349206
GRAMERCY	0.05128205	0.07692308	0.06153846
GREENWICH VILLAGE-CENTRAL	0.00000000	0.00000000	NaN
GREENWICH VILLAGE-WEST	0.16981132	0.17307692	0.17142857
HARLEM-CENTRAL	0.35510204	0.29000000	0.31926606
HARLEM-EAST	0.05970149	0.09302326	0.07272727
HARLEM-UPPER	0.04615385	0.04838710	0.04724409
HARLEM-WEST	0.15384615	0.28571429	0.20000000
INWOOD	0.05555556	0.33333333	0.09523810
JAVITS CENTER	0.00000000	0.00000000	NaN
KIPS BAY	0.11320755	0.07894737	0.09302326
LITTLE ITALY	0.00000000	0.00000000	NaN
LOWER EAST SIDE	0.04761905	0.04761905	0.04761905

MANHATTAN-UNKNOWN	NaN	NaN	NaN
MANHATTAN VALLEY	0.10638298	0.11627907	0.11111111
MIDTOWN CBD	0.00000000	0.00000000	NaN
MIDTOWN EAST	0.08219178	0.08333333	0.08275862
MIDTOWN WEST	0.08333333	0.06896552	0.07547170
MORNINGSIDE HEIGHTS	NaN	NaN	NaN
MURRAY HILL	0.01886792	0.02083333	0.01980198
ROOSEVELT ISLAND	0.00000000	0.00000000	NaN
SOHO	0.03846154	0.04545455	0.04166667
SOUTHBRIDGE	0.00000000	0.00000000	NaN
TRIBECA	0.07692308	0.06493506	0.07042254
UPPER EAST SIDE (59-79)	0.09900990	0.08771930	0.09302326
UPPER EAST SIDE (79-96)	0.05504587	0.05714286	0.05607477
UPPER EAST SIDE (96-110)	0.00000000	0.00000000	NaN
UPPER WEST SIDE (59-79)	0.07692308	0.06730769	0.07179487
UPPER WEST SIDE (79-96)	0.09523810	0.08695652	0.09090909
UPPER WEST SIDE (96-116)	0.24242424	0.33333333	0.28070175
WASHINGTON HEIGHTS LOWER	0.12820513	0.18518519	0.15151515
WASHINGTON HEIGHTS UPPER	0.06250000	0.07692308	0.06896552

Random Forest Model

```
> data.frame(recall, precision, f1)
```

	recall	precision	f1
ALPHABET CITY	0.9583333	1.0000000	0.9787234
CHELSEA	0.9718310	0.9718310	0.9718310
CHINATOWN	0.8947368	0.9444444	0.9189189
CIVIC CENTER	0.9230769	0.8571429	0.8888889
CLINTON	1.0000000	0.9803922	0.9900990
EAST VILLAGE	0.9791667	0.9400000	0.9591837
FASHION	0.9117647	1.0000000	0.9538462
FINANCIAL	0.9838710	1.0000000	0.9918699
FLATIRON	1.0000000	0.9705882	0.9850746
GRAMERCY	1.0000000	1.0000000	1.0000000
GREENWICH VILLAGE-CENTRAL	0.8695652	0.9090909	0.8888889
GREENWICH VILLAGE-WEST	1.0000000	0.9636364	0.9814815
HARLEM-CENTRAL	0.9673469	0.9793388	0.9733060
HARLEM-EAST	0.9552239	0.9696970	0.9624060
HARLEM-UPPER	0.9846154	0.9411765	0.9624060
HARLEM-WEST	0.9230769	0.9230769	0.9230769
INWOOD	1.0000000	0.9473684	0.9729730
JAVITS CENTER	1.0000000	1.0000000	1.0000000
KIPS BAY	1.0000000	1.0000000	1.0000000
LITTLE ITALY	1.0000000	0.7500000	0.8571429
LOWER EAST SIDE	0.9365079	0.9516129	0.9440000
MANHATTAN-UNKNOWN	NaN	NaN	NaN
MANHATTAN VALLEY	0.9574468	0.9375000	0.9473684
MIDTOWN CBD	1.0000000	1.0000000	1.0000000
MIDTOWN EAST	1.0000000	1.0000000	1.0000000
MIDTOWN WEST	0.9895833	1.0000000	0.9947644

MORNINGSIDE HEIGHTS	NaN	NaN	NaN
MURRAY HILL	1.0000000	1.0000000	1.0000000
ROOSEVELT ISLAND	1.0000000	1.0000000	1.0000000
SOHO	0.9230769	0.9230769	0.9230769
SOUTHBRIDGE	1.0000000	0.8888889	0.9411765
TRIBECA	0.9384615	0.9682540	0.9531250
UPPER EAST SIDE (59-79)	0.9801980	1.0000000	0.9900000
UPPER EAST SIDE (79-96)	1.0000000	0.9732143	0.9864253
UPPER EAST SIDE (96-110)	0.3333333	1.0000000	0.5000000
UPPER WEST SIDE (59-79)	0.9890110	1.0000000	0.9944751
UPPER WEST SIDE (79-96)	1.0000000	0.9692308	0.9843750
UPPER WEST SIDE (96-116)	1.0000000	1.0000000	1.0000000
WASHINGTON HEIGHTS LOWER	1.0000000	1.0000000	1.0000000
WASHINGTON HEIGHTS UPPER	0.9375000	1.0000000	0.9677419

Decision Tree Model

```
> data.frame(recall, precision, f1)
```

	recall	precision	f1
ALPHABET CITY	0.0000000	NaN	NaN
CHELSEA	0.0000000	NaN	NaN
CHINATOWN	0.0000000	NaN	NaN
CIVIC CENTER	0.0000000	NaN	NaN
CLINTON	0.0000000	NaN	NaN
EAST VILLAGE	0.0000000	NaN	NaN
FASHION	0.0000000	NaN	NaN
FINANCIAL	0.0000000	NaN	NaN
FLATIRON	0.0000000	NaN	NaN
GRAMERCY	0.0000000	NaN	NaN
GREENWICH VILLAGE-CENTRAL	0.0000000	NaN	NaN
GREENWICH VILLAGE-WEST	0.0000000	NaN	NaN
HARLEM-CENTRAL	0.9795918	0.56206089	0.7142857
HARLEM-EAST	0.0000000	NaN	NaN
HARLEM-UPPER	0.0000000	NaN	NaN
HARLEM-WEST	0.0000000	NaN	NaN
INWOOD	0.0000000	NaN	NaN
JAVITS CENTER	0.0000000	NaN	NaN
KIPS BAY	0.0000000	NaN	NaN
LITTLE ITALY	0.0000000	NaN	NaN
LOWER EAST SIDE	0.0000000	NaN	NaN
MANHATTAN-UNKNOWN	NaN	NaN	NaN
MANHATTAN VALLEY	0.0000000	NaN	NaN
MIDTOWN CBD	0.0000000	NaN	NaN
MIDTOWN EAST	0.0000000	NaN	NaN
MIDTOWN WEST	1.0000000	0.07830343	0.1452345
MORNINGSIDE HEIGHTS	NaN	NaN	NaN
MURRAY HILL	0.0000000	NaN	NaN
ROOSEVELT ISLAND	0.0000000	NaN	NaN
SOHO	0.0000000	NaN	NaN
SOUTHBRIDGE	0.0000000	NaN	NaN

TRIBECA	0.0000000	NaN	NaN
UPPER EAST SIDE (59-79)	0.0000000	NaN	NaN
UPPER EAST SIDE (79-96)	0.9541284	0.52000000	0.6731392
UPPER EAST SIDE (96-110)	0.0000000	NaN	NaN
UPPER WEST SIDE (59-79)	0.0000000	NaN	NaN
UPPER WEST SIDE (79-96)	0.0000000	NaN	NaN
UPPER WEST SIDE (96-116)	0.0000000	NaN	NaN
WASHINGTON HEIGHTS LOWER	0.0000000	NaN	NaN
WASHINGTON HEIGHTS UPPER	0.0000000	NaN	NaN

2.

- a. I applied the same univariate models (especially GROSS.SQUARE.FEET and YEAR.BUILT) to the Bronx data: in the Bronx the univariate regression on GROSS.SQUARE.FEET had a higher R^2 (~0.0448) and was highly significant ($p < 2e-16$), and YEAR.BUILT also showed significance though with small R^2 , similar to Manhattan. Residual-vs-fitted plots (which you produced) show heteroskedasticity and remaining large residuals, but overall the GROSS.SQUARE.FEET relationship explains slightly more variance in Bronx than Manhattan. This suggests the GSF $\rightarrow$ price relationship generalizes somewhat, but absolute predictive power remains low because many other features (neighborhood-specific demand, building condition) are still missing; the better R^2 in Bronx may reflect a more homogeneous market or fewer extreme ultra-high prices.

```
> for (feature in features) {
+   formula <- as.formula(paste("SALE.PRICE ~", feature))
+   lin.mod <- lm(formula, data = bronx_data)
+
+   print(feature)
+   print(summary(lin.mod))
+
+   # prepare data for ggplot
+   df <- data.frame(
+     fitted = lin.mod$fitted.values,
+     resid = lin.mod$residuals
+   )
+
+   # residual plot
+   print(
+     ggplot(df, aes(x = fitted, y = resid)) +
```



```
+ geom_point() +
+ geom_hline(yintercept = 0, linetype = "dashed") +
+ labs(title = paste0("Residuals vs Fitted (", feature, ")"))
+ )
+ }
[1] "YEAR.BUILT"
```

Call:

```
lm(formula = formula, data = bronx_data)
```

Residuals:

	Min	1Q	Median	3Q	Max
	-535210	-154954	10322	173756	875605

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	-1.016e+06	1.381e+05	-7.352	2.04e-13 ***
YEAR.BUILT	7.697e+02	7.117e+01	10.814	< 2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 276700 on 16501 degrees of freedom

Multiple R-squared: 0.007037, Adjusted R-squared: 0.006977

F-statistic: 116.9 on 1 and 16501 DF, p-value: < 2.2e-16

[1] "SALE.PRICE"

Call:

```
lm(formula = formula, data = bronx_data)
```

Residuals:

	Min	1Q	Median	3Q	Max
	-478014	-153015	10985	171985	856985

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	478015	2162	221.1	<2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 277700 on 16502 degrees of freedom

[1] "Latitude"

Call:

```
lm(formula = formula, data = bronx_data)
```

Residuals:

	Min	1Q	Median	3Q	Max
	-478851	-153668	11353	171570	856527

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	229133	3333628	0.069	0.945
Latitude	6104	81600	0.075	0.940

Residual standard error: 277200 on 16040 degrees of freedom

(461 observations deleted due to missingness)

Multiple R-squared: 3.489e-07, Adjusted R-squared: -6.2e-05

F-statistic: 0.005596 on 1 and 16040 DF, p-value: 0.9404

[1] "Longitude"

Call:

lm(formula = formula, data = bronx_data)

Residuals:

	Min	1Q	Median	3Q	Max
	-484087	-154245	10808	172022	852996

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	-5671795	5508884	-1.030	0.303
Longitude	-83269	74585	-1.116	0.264

Residual standard error: 277200 on 16040 degrees of freedom

(461 observations deleted due to missingness)

Multiple R-squared: 7.77e-05, Adjusted R-squared: 1.536e-05

F-statistic: 1.246 on 1 and 16040 DF, p-value: 0.2643

[1] "GROSS.SQUARE.FEET"

Call:

lm(formula = formula, data = bronx_data)

Residuals:

	Min	1Q	Median	3Q	Max
	-1004578	-137570	26126	170466	852583

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	3.535e+05	4.950e+03	71.40	<2e-16 ***
GROSS.SQUARE.FEET	5.385e+01	1.936e+00	27.82	<2e-16 ***

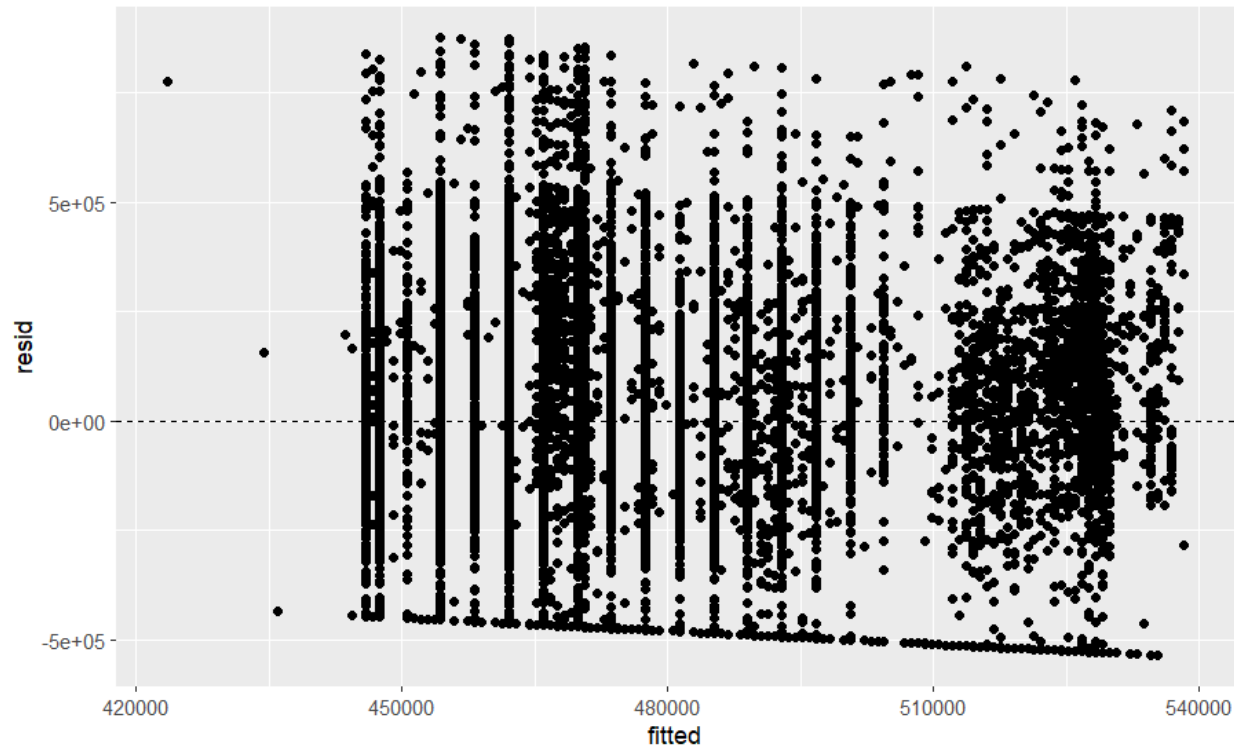
Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 271400 on 16501 degrees of freedom

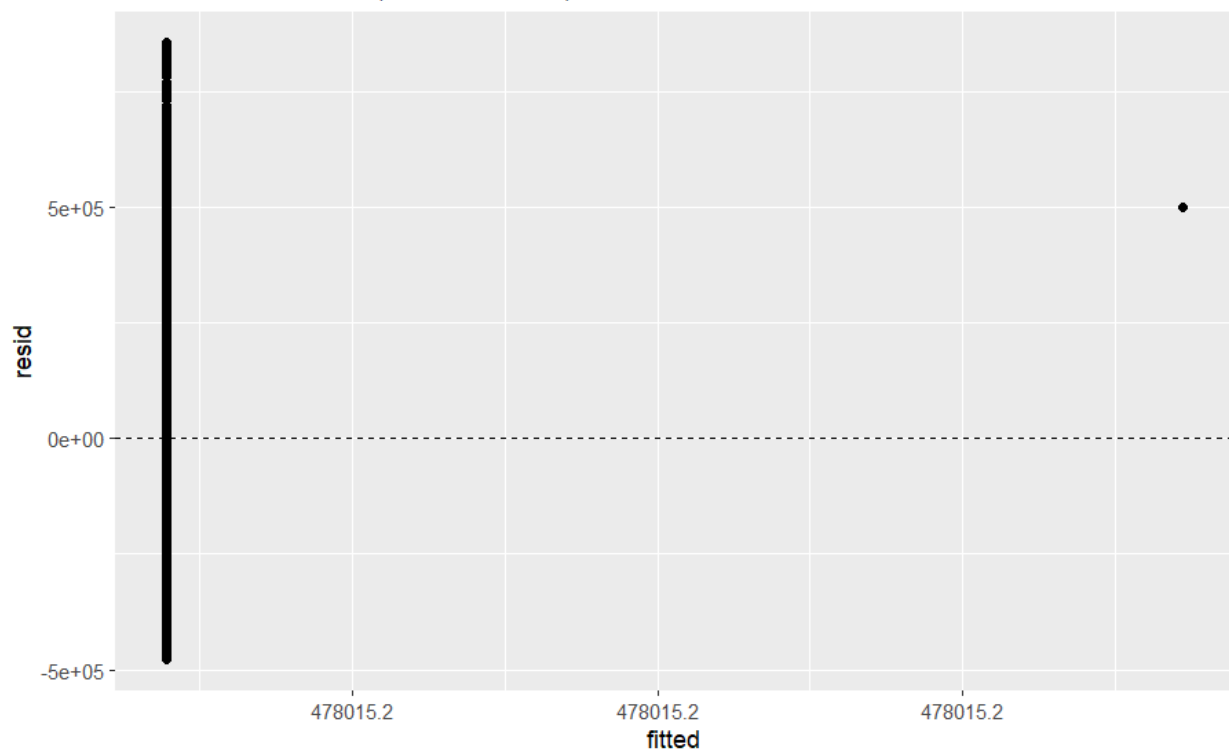
Multiple R-squared: 0.0448, Adjusted R-squared: 0.04474

F-statistic: 773.8 on 1 and 16501 DF, p-value: $< 2.2e-16$

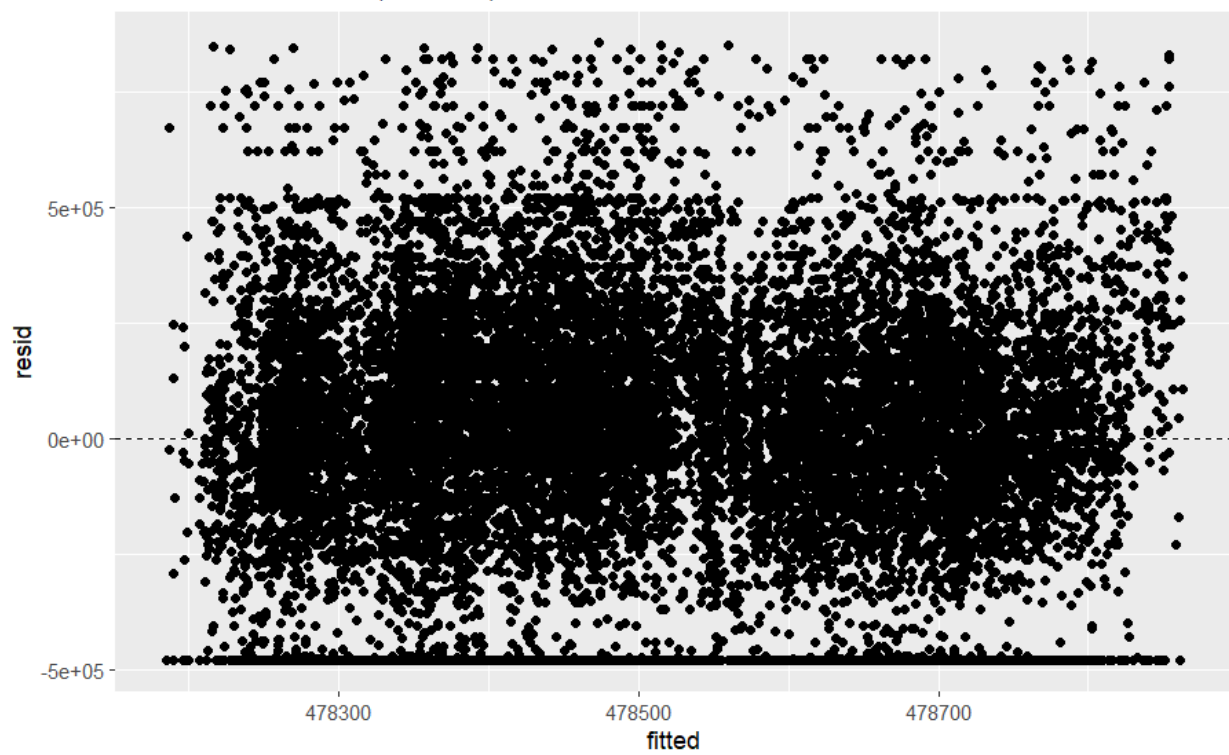
Residuals vs Fitted (YEAR.BUILT)



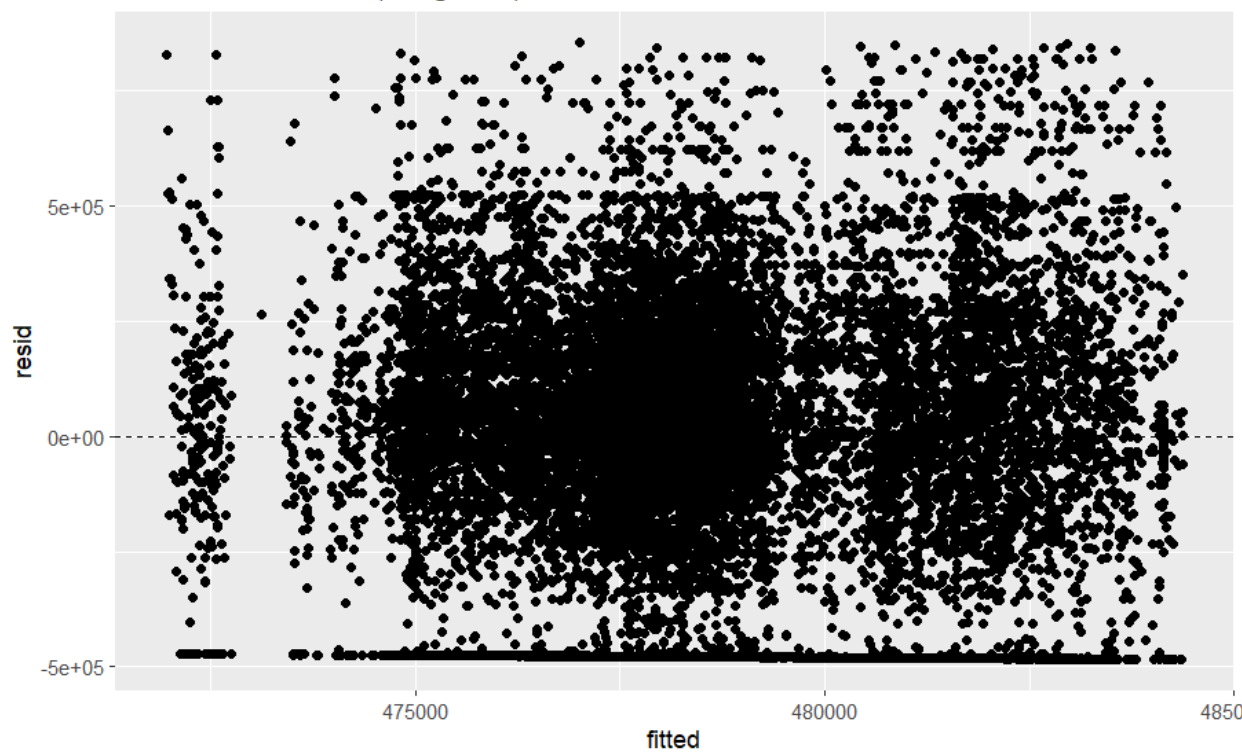
Residuals vs Fitted (SALE.PRICE)



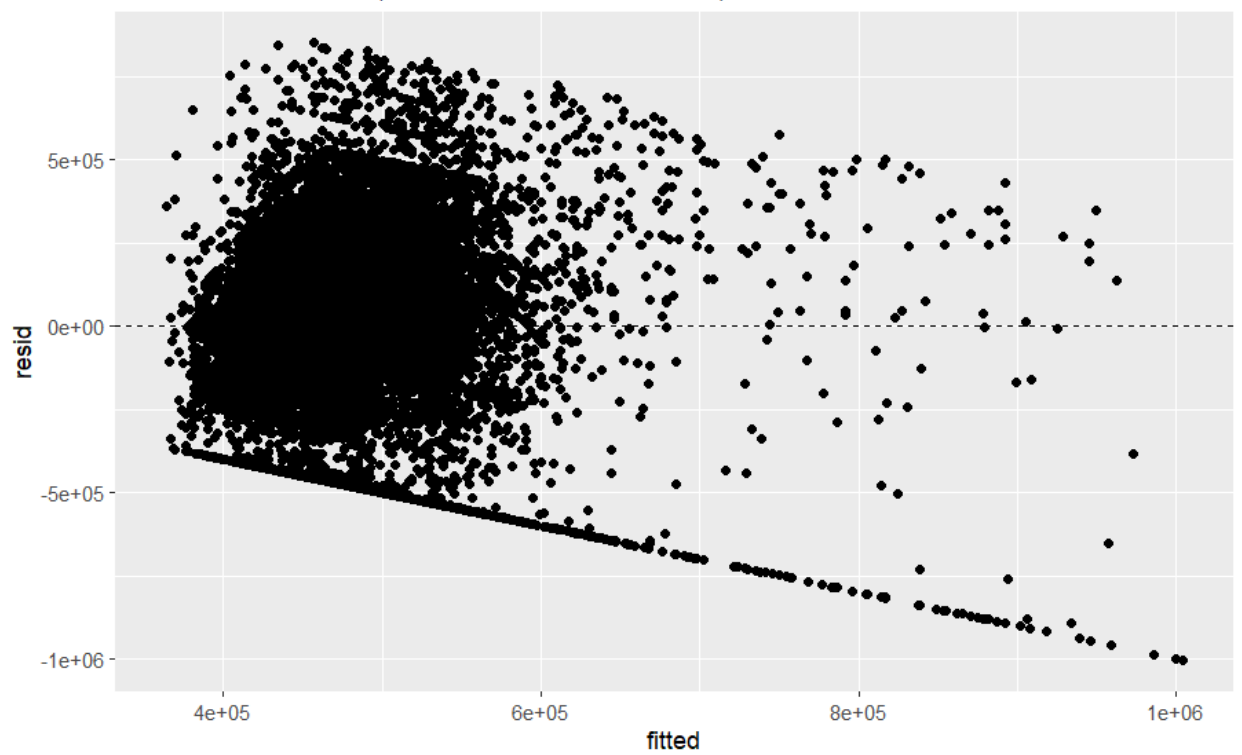
Residuals vs Fitted (Latitude)



Residuals vs Fitted (Longitude)



Residuals vs Fitted (GROSS.SQUARE.FEET)



- b. When applying classification models trained on Manhattan to Bronx, performance drops markedly: the Random Forest you ran on Bronx produced much higher per-class metrics when trained/tested on Bronx itself (many near 1.0), but cross-borough generalization (train on Manhattan → test Bronx) performs poorly because neighborhood labels do not match and features' distributions shift. In particular, class definitions, class priors, and feature-label relationships differ across boroughs, so a model trained on Manhattan neighborhoods cannot reliably predict Bronx neighborhoods; this is reflected in low recall/precision for many classes under cross-borough testing. The primary causes are covariate shift (different price and area distributions) and label mismatch (different neighborhood taxonomy).

```
> data.frame(recall, precision, f1)
```

	recall	precision	f1
BATHGATE	0.000000000	0.000000000	NaN
BAYCHESTER	0.156177156	0.10323575	0.12430427
BEDFORD PARK/NORWOOD	0.009259259	0.01369863	0.01104972
BELMONT	0.019230769	0.033333333	0.02439024
BRONX-UNKNOWN	0.000000000	NaN	NaN
BRONX PARK	0.000000000	0.000000000	NaN
BRONXDALE	0.065134100	0.06882591	0.06692913
CASTLE HILL/UNIONPORT	0.055555556	0.06870229	0.06143345
CITY ISLAND	0.027397260	0.05000000	0.03539823
CITY ISLAND-PELHAM STRIP	NaN	0.000000000	NaN
CO-OP CITY	0.000000000	NaN	NaN
COUNTRY CLUB	0.054644809	0.05235602	0.05347594
CROTONA PARK	0.000000000	0.000000000	NaN
EAST TREMONT	0.011494253	0.02564103	0.01587302
FIELDSTON	0.142857143	0.14285714	0.14285714
FORDHAM	0.018518519	0.04761905	0.02666667
HIGHBRIDGE/MORRIS HEIGHTS	0.045454545	0.06060606	0.05194805
HUNTS POINT	0.024390244	0.04545455	0.03174603
KINGSBRIDGE HTS/UNIV HTS	0.000000000	0.000000000	NaN
KINGSBRIDGE/JEROME PARK	0.000000000	0.000000000	NaN
MELROSE/CONCOURSE	0.020833333	0.04081633	0.02758621
MORRIS PARK/VAN NEST	0.064655172	0.06000000	0.06224066
MORRISANIA/LONGWOOD	0.102222222	0.10087719	0.10154525
MOTT HAVEN/PORT MORRIS	0.056179775	0.07812500	0.06535948
MOUNT HOPE/MOUNT EDEN	0.019607843	0.04166667	0.02666667
PARKCHESTER	0.009523810	0.01639344	0.01204819

PELHAM BAY	NaN	NaN	NaN
PELHAM GARDENS	0.016129032	0.02702703	0.02020202
PELHAM PARKWAY NORTH	0.023255814	0.02970297	0.02608696
PELHAM PARKWAY SOUTH	0.000000000	0.00000000	NaN
RIVERDALE	0.043478261	0.04395604	0.04371585
SCHUYLERVILLE/PELHAM BAY	0.080168776	0.07786885	0.07900208
SOUNDVIEW	0.171428571	0.12587413	0.14516129
THROGS NECK	0.120845921	0.10000000	0.10943912
WAKEFIELD	0.083003953	0.07142857	0.07678245
WESTCHESTER	0.008547009	0.01369863	0.01052632
WILLIAMSBRIDGE	0.156108597	0.11577181	0.13294798
WOODLAWN	0.000000000	0.00000000	NaN

```
> data.frame(recall, precision, f1)
```

	recall	precision	f1
BATHGATE	0.9818182	1.0000000	0.9908257
BAYCHESTER	0.9976690	0.9907407	0.9941928
BEDFORD PARK/NORWOOD	0.9907407	0.9907407	0.9907407
BELMONT	0.9807692	1.0000000	0.9902913
BRONX-UNKNOWN	0.3333333	1.0000000	0.5000000
BRONX PARK	0.0000000	NaN	NaN
BRONXDALE	0.9770115	0.9922179	0.9845560
CASTLE HILL/UNIONPORT	0.9938272	1.0000000	0.9969040
CITY ISLAND	1.0000000	1.0000000	1.0000000
CITY ISLAND-PELHAM STRIP	NaN	NaN	NaN
CO-OP CITY	0.0000000	NaN	NaN
COUNTRY CLUB	1.0000000	0.9891892	0.9945652
CROTONA PARK	1.0000000	0.9787234	0.9892473
EAST TREMONT	0.9885057	0.9885057	0.9885057
FIELDSTON	0.8571429	0.7058824	0.7741935
FORDHAM	0.9629630	0.9454545	0.9541284
HIGHBRIDGE/MORRIS HEIGHTS	0.9886364	0.9775281	0.9830508
HUNTS POINT	0.9024390	1.0000000	0.9487179
KINGSBRIDGE HTS/UNIV HTS	0.9843750	0.9402985	0.9618321
KINGSBRIDGE/JEROME PARK	0.9583333	1.0000000	0.9787234
MELROSE/CONCOURSE	0.9791667	0.9791667	0.9791667
MORRIS PARK/VAN NEST	0.9956897	0.9665272	0.9808917
MORRISANIA/LONGWOOD	0.9822222	0.9778761	0.9800443
MOTT HAVEN/PORT MORRIS	0.9887640	0.9887640	0.9887640
MOUNT HOPE/MOUNT EDEN	0.9803922	0.9803922	0.9803922
PARKCHESTER	0.9523810	0.9803922	0.9661836
PELHAM BAY	NaN	NaN	NaN
PELHAM GARDENS	1.0000000	0.9841270	0.9920000
PELHAM PARKWAY NORTH	0.9767442	1.0000000	0.9882353
PELHAM PARKWAY SOUTH	0.9482759	0.9821429	0.9649123
RIVERDALE	0.9565217	0.9777778	0.9670330
SCHUYLERVILLE/PELHAM BAY	1.0000000	0.9916318	0.9957983
SOUNDVIEW	1.0000000	0.9952607	0.9976247
THROGS NECK	0.9909366	1.0000000	0.9954476

WAKEFIELD	0.9960474	1.0000000	0.9980198
WESTCHESTER	0.9572650	0.9572650	0.9572650
WILLIAMSBRIDGE	1.0000000	0.9888143	0.9943757
WOODLAWN	1.0000000	1.0000000	1.0000000

```
> data.frame(recall, precision, f1)
```

	recall	precision	f1
BATHGATE	0.0000000	NaN	NaN
BAYCHESTER	0.8904429	0.6574871	0.7564356
BEDFORD PARK/NORWOOD	0.0000000	NaN	NaN
BELMONT	0.0000000	NaN	NaN
BRONX-UNKNOWN	0.0000000	NaN	NaN
BRONX PARK	0.0000000	NaN	NaN
BRONXDALE	0.0000000	NaN	NaN
CASTLE HILL/UNIONPORT	0.0000000	NaN	NaN
CITY ISLAND	0.0000000	NaN	NaN
CITY ISLAND-PELHAM STRIP	NaN	NaN	NaN
CO-OP CITY	0.0000000	NaN	NaN
COUNTRY CLUB	0.0000000	NaN	NaN
CROTONA PARK	0.0000000	NaN	NaN
EAST TREMONT	0.0000000	NaN	NaN
FIELDSTON	0.0000000	NaN	NaN
FORDHAM	0.0000000	NaN	NaN
HIGHBRIDGE/MORRIS HEIGHTS	0.0000000	NaN	NaN
HUNTS POINT	0.0000000	NaN	NaN
KINGSBRIDGE HTS/UNIV HTS	0.0000000	NaN	NaN
KINGSBRIDGE/JEROME PARK	0.0000000	NaN	NaN
MELROSE/CONCOURSE	0.0000000	NaN	NaN
MORRIS PARK/VAN NEST	0.0000000	NaN	NaN
MORRISANIA/LONGWOOD	0.0000000	NaN	NaN
MOTT HAVEN/PORT MORRIS	0.0000000	NaN	NaN
MOUNT HOPE/MOUNT EDEN	0.0000000	NaN	NaN
PARKCHESTER	0.0000000	NaN	NaN
PELHAM BAY	NaN	NaN	NaN
PELHAM GARDENS	0.0000000	NaN	NaN
PELHAM PARKWAY NORTH	0.0000000	NaN	NaN
PELHAM PARKWAY SOUTH	0.0000000	NaN	NaN
RIVERDALE	0.0000000	NaN	NaN
SCHUYLerville/PELHAM BAY	0.0000000	NaN	NaN
SOUNDVIEW	1.0000000	0.1366298	0.2404121
THROGS NECK	0.0000000	NaN	NaN
WAKEFIELD	0.0000000	NaN	NaN
WESTCHESTER	0.0000000	NaN	NaN
WILLIAMSBRIDGE	0.9796380	0.3798246	0.5474083
WOODLAWN	0.0000000	NaN	NaN

- c. Overall, the strongest single quantitative predictor across boroughs was GROSS.SQUARE.FEET, but even that explains only a small share

of variance; classification works well within a borough (especially with Random Forest) but fails to generalize across boroughs because neighborhoods and price/area distributions differ. I'm moderately confident in the within-borough classification results and cautious about cross-borough regression/classification generalization unless more features (condition, building class, sale type) or borough-specific re-training are used.